Final Submission

Project Information

Project ID: CMPG325-2026-072

Client ID: CLI-072

MENGWAI, KKE

Student Number: 44419465

Submission Date: 16 October 2026

Submitted Deliverables

1. Packet Tracer Project File

   Location: 04_Packet_Tracer/CMPG325_Final_MENGWAI.pkt

   Description: Fully working network with VLSM, inter-VLAN routing, DHCP, and ACL

2. GitHub Portfolio of Evidence

   Location: This repository

   Description: Complete record of planning, design, implementation, testing, and reflection

3. Technical Report

   - Location: TECHNICAL_REPORT.pdf
   - Description: Full technical documentation of the project

4. Video Demonstration

   - Location: 06_Video_Demonstration/VIDEO_LINK.md
   - Description: 15-20 minute video with inset webcam

5. Additional Files

   a) All testing screenshots in 05_Testing_Evidence/
   b) All configuration files

Summary of Implementation

- VLSM: 8 subnets allocated from 172.30.46.0/23

- VLANs: 8 VLANs configured (10, 20, 30, 40, 50, 60, 70, 99)
- Routing: Inter-VLAN routing via router-on-a-stick
- ACL: Extended ACL on Edge Router ensuring Management always has internet
- CR2: Additional floor fully integrated (VLAN 70)

Testing Summary

Test	Result
Management internet access	Pass
Staff internet blocked	Pass
Inter-VLAN communication	Pass
CR2 connectivity	Pass
Routing table correct	Pass
VLAN assignments correct	Pass
ACL match counts	Pass

Declaration

I, MENGWAI, KKE (44419465), confirm that this submission is my own original work and complies with the NWU AI Policy and the CMPG 325 Academic Integrity guidelines.

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